Strategic Electoral Mobility*

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August 2026

Abstract

Equal political representation is a principle both widely praised and often violated. Votes in many democratic systems do in fact carry more weight in some places than in others. This paper, using the case of Poland – where substantial vote power disparities are coupled with unconstrained relocations across districts – studies electoral mobility governed by pivotality choice. Through a theoretical model and using municipality-level electoral data, we show that voters select their voting location strategically to maximize their ballot's weight, and study mobilization among local voters. A counterfactual with equal vote weight across the country estimates above 30,000 voters relocating strategically. Local turnout response is consistent with local mobilization in municipalities exposed to inflows.

1 Introduction

In most democratic systems, electoral rules are designed in advance, so that on the ballot day voters take them as given, no matter how unfair they would be: voters either vote for supported candidates or abstain. If political competition in a district is negligible or the number of apportioned seats does not reflect the size of local population, a vote cast elsewhere would count *more*. Nothing reveals such disparities

*The paper received (ex-aequo) the 2026 Jacques Rupnik Prize for the best PhD paper presented at the EUI Workshop on Political Behaviour in Central and Eastern Europe. We would like to express gratitude for generous comments to Giacomo Calzolari, Zeinab Aboutalebi, Yusuf Agus, Enrico Cantoni, Christian Cox, Marta Kołczyńska, Alberto Venturin, Monika Wesołowska and numerous participants of the Political Behaviour Colloquium and the BDC Working Group at the EUI. All mistakes are our own.

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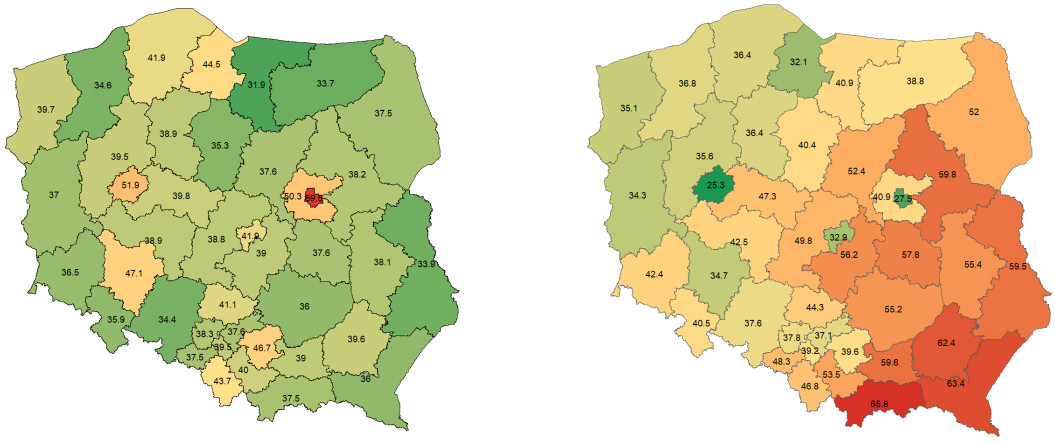
in vote weight as clearly as campaign spending: in the 2024 US presidential election, 70% of all funds was spent in only as much as seven key states (NPR, 2024). Were voters able to select their voting location based on perceived impact, one could expect such imbalances to gradually decline. If such electoral disparity disproportionately affects supporters of a particular party, the problem also becomes one of coordination: enough voters should optimally leave a disadvantaged district to make a difference elsewhere, without losing it at the same time. Can voters successfully counteract adverse institutional framework that distorts equality of representation without changing electoral rules?

This paper studies a rare setting where voters can in fact choose the constituency they vote in. Poland's 2023 parliamentary election provides an example of such electoral mobility. Two institutional features made strategic re-registration both feasible and enticing: malapportionment – unequal representation arising from differences in the number of voters per seat across constituencies – and limited administrative barriers to a temporary change of voting location. Together, these features allowed voters to redirect their ballots to constituencies where their vote carried more weight.

Polish electoral system is designed to approximate equal representation across voters regardless of where they cast their ballot. The number of seats assigned to each of 41 lower-chamber (Sejm) constituencies is intended to proportionally reflect the size of population in the constituency. However, because the system does not include an automatic population-based reapportionment mechanism, changes in the allocation of seats require a parliamentary decision. The last time it happened was ahead of 2011 elections. Combined with substantial internal migration, especially from rural areas to large cities, this has generated persistent malapportionment. By the 2019 parliamentary elections, a vote in some depopulating rural districts carried almost twice as much weight as a vote cast in Warsaw (Figure 1, left panel). At the same time, constituencies with the highest number of voters per seat – and therefore the weakest vote power – coincided with districts in which support for the incumbent Law and Justice (PiS) party was lowest (Figure 1, right panel). As a result, malapportionment systematically disadvantaged urban voters, among whom opposition parties were particularly popular.

The second institutional feature makes this distortion strategically actionable. Polish voters can change the location of their vote to another constituency, effectively allowing them to cast a ballot in a district where their vote may be more consequential. The re-registration procedure was designed to accommodate voters travelling on

election day[1], as well as those who had moved within the country without updating their permanent registration. Re-registration does not require a minimum residence period or a justification. It can be done in two ways: (i) an online application in which voters indicate the location where they wish to vote; or (ii) a voting certificate, collected from a municipal office, which gives voters full flexibility over their voting location on election day.



(a) Vote power across constituencies

(b) Law and Justice support

Figure 1: Malapportionment and party support across electoral constituencies. Panel (a) shows the number of voters, in thousands, per seat across constituencies in 2019. Panel (b) shows support for Law and Justice (PiS) in the 2019 parliamentary election.

This strategic opportunity became politically salient during the 2023 parliamentary election, held on October 15. The political scene was highly polarized after eight years of the Law and Justice government, during which international organizations documented a deterioration in democratic governance and the rule of law[2]. For this reason, the 2023 election was widely seen as one of the most consequential Polish elections since the semi-free election of 1989. The main opposition parties – Civic Coalition, Third Way, and the New Left – were often grouped together as the democratic opposition and were strongest in urban areas. In the run-up to the election, many voters became aware of the persistent disadvantage associated with voting in large cities. Re-registration created an opportunity for a strategic electoral mobility

[1]General elections in Poland are always held on Sundays.

[2]https://freedomhouse.org/country/poland/nations-transit/2023

on the election day to increase the impact of a vote. Despite limited official encouragement from political parties, grassroots attempts to coordinate emerged online. A number of online guides prepared by individual voters used data from 2019 parliamentary elections and current opinion polls to provide information where to vote to maximize ballot's impact, based on party preferences and a place of origin[3]. Google Trends data indicates that searches related to changing voting location rose sharply before the 2023 election, peaking far above comparable searches prior to the 2019 election (see Figure 5 in the Appendix).

The availability of strategic electoral mobility reshaped the traditional voting decision, usually limited to the choice between voting and abstaining. In the considered setting, highly motivated voters faced instead a two-stage decision:

1. Should I vote?

2. Where should I vote to maximize the impact of my ballot?

This introduced an individual level cost-benefit calculation trading off the effort and impact of traveling to a different constituency. It also created a rare case in which voting location became an endogenous choice driven by perceived impact. In this paper, we first develop a theoretical framework which describes and aggregates individual voting decisions in the context of potential strategic mobility. Then, using data from the 2023 Polish parliamentary election we ask whether these incentives and the sum of individual choices translated into large-scale strategic re-registration across constituencies. Finally, we examine whether the resulting flows affected turnout of local voters in the locations exposed to voters' mobility. The paper therefore addresses two research questions:

1. Did pivotality differences between constituencies lead to significant strategic relocation of voters?

2. What were the implications of these flows for turnout in the affected constituencies?

The paper makes the following two main contributions. First, it estimates a voter-side response to malapportionment. While unequal representation is usually an institutional constraint faced passively by voters, Polish parliamentary election setting

[3]Most of these guides are, unfortunately, no longer available online. One of them can be still accessed under: https://github.com/ptrstpp950/wybieramTakMapa (access on 27.08.2026).

allowed voters to respond actively by strategically changing the location of their ballot. Using municipality-level electoral data, we provide evidence that a significant number of voters re-registered for strategic reasons, moving their votes toward constituencies where their marginal impact was higher. Our estimations yield around 33,000 strategically relocated voters.

Second, the paper studies equilibrium response of non-movers to the strategic mobility of some voters. By estimating the change in turnout in the constituencies with high strategic inflows and outflows relative to their size, we examine whether strategic relocations affected participation of local population compared to the previous elections. This connects institutional discrepancies in the marginal impact of vote to voters' location choices and subsequent local mobilization. Our results are consistent with mild local mobilization patterns. Given past turnout levels and trends, municipalities with higher exposure to strategic inflows experienced faster turnout growth than otherwise expected, also having accounted for the direct effect of incoming voters.

The remainder of the paper is organized as follows. Section 2 provides a literature review. Section 3 develops a theoretical framework that delivers predictions about pivotality, the distribution of mobility costs, and equilibrium voting decisions. Section 4 contains the empirical analysis of the size of strategic mobility and its equilibrium turnout effects. Concluding remarks close the paper.

2 Literature review

This paper is the first to study uneven vote power, voters' mobility and the resulting turnout effects jointly, adding to several literatures within political economy.

Both the proportional representation and the first-past-the-post electoral systems often deviate from the principle of equal representation, as documented in (Samuels and Snyder, 2001) and (Beramendi, Boix, Guinjoan, and Rogers, 2022). So-called malapportionment, i.e., uneven ratio between the size of a constituency's population and the number of its elected representatives, may arise at different stages of the seats assignment process (decomposed in (Kamahara, Wada, and Kasuya, 2021)), or be a result of a strategic choice of the incumbents. Too few or too many seats assigned to a district relative to its population or redrawing its borders in a way that favors some candidates give unequal leverage on the election's results to some areas at the expense of others. Riera and Lago (2023) find that this effect is particularly

strong when incumbents possess reliable information about the spatial distribution of the partisan support. Several other studies indicate that malapportionment tends to be larger in newly democratized countries: (Samuels and Snyder, 2001), (Kamahara, Wada, and Kasuya, 2021), (Beramendi, Boix, Guinjoan, and Rogers, 2022). Sometimes, malapportionment may come from a mixture of strategic factors, institutional rigidity, and exogenous social changes. In Poland, outdated seat allocation (unchanged since 2011) has effectively increased vote power in rapidly depopulating rural areas: (Mazur and Gendźwiłł, 2024), (Zubik and Kaczmarczyk, 2025). Using the malapportionment index from (Samuels and Snyder, 2001), between 1997 and 2023 Poland quadrupled its vote power disparity from around 1.7% to almost 6.5%. The index[4] measures the share of seats elected in districts other than those they would be assigned to under equal representation. With a sample mean at 7% for almost 80 considered lower chambers in (Samuels and Snyder, 2001), it shows a widespread character of malapportionment across democratic systems.

Malapportionment is often associated with an advantage to rural districts and thus to more conservative parties. Boone and Wahman (2015) argue that in many African countries it was an institutional design of pre-democratic incumbents, while Johnson and Miller (2023) document the size of the Republican bias in the US Senate, stemming from the same representation of each state regardless of its population. However, some recent studies like (Haas, Miller, and Kimbrough, 2022) and (Walter and Emmenegger, 2025) also point out that urban or minority-dominated districts may benefit from population-based seat assignments when only total population is considered. If parties maximize their winning chances by implementing policies that benefit their core political constituencies (as in the classical partisan model of Hibbs Jr (1977)), malapportionment will also affect policymaking. Overall, it is documented to have undesired economic effects. It diverts funding from underrepresented constituencies (Ansolabehere, Gerber, and Snyder, 2002), slows down development (Hiroi, 2019), (Bhavnani, 2021), and is associated with higher inequalities (Beramendi, Rogers, and Díaz-Cayeros, 2017) and lower tax revenues (Ardanaz and Scartascini, 2013).

In most cases voters can do little to counteract their decreasing vote power. Some related works do, however, indicate that grassroots mobilization can effectively weaken the effects of policies aimed at reducing some groups' willingness to participate, such as stricter ID laws (Cantoni and Pons, 2021), (Cantoni, Pons, and Schafer, 2025).

[4]Half of the sum of absolute differences between the shares of total seats to shares of total population, per district.

A few documented cases, such as American elections of 2000 (Earl and Schussman, 2004) or British elections of 2019 (Nicholls and Hayton, 2020) provide evidence of cross-district coordination of voters backed by Internet sites. Without changing voting location, voters were agreeing to swap who they cast vote for, given expected local support of similar candidates. These cases have, however, received only little empirical examination. At the same time, political concerns may be one of determinants of a residence, and thus also a legislature, choice, leading to spatial sorting (Gimpel and Hui, 2015), though only of a limited strength, as argued by Mummolo and Nall (2017).

Standard approaches to voters' coordination focus instead on within-district coordination across different candidates, as in (Fey, 1997), (Deltas, Herrera, and Polborn, 2016), and (Granzier, Pons, and Tricaud, 2023). In the case when the highest-support candidate (so called Condorcet loser) is placed at the bottom of personal rankings by the majority of voters potentially split between several other contesters, strategic voting can secure socially optimal outcomes. Theoretical works of Fey (1997) and Andonie and Diermeier (2017) consider implementability and stability of coordinated equilibria supported by pre-electoral polls. In empirical studies, Forsythe and Myerson (1991) and Hall and Snyder (2015) also find that voters use this type of information to coordinate on expectedly more successful candidates. Similar effect can be produced by sequential voting, which amplifies learning, though at a risk of coordinating on a lower-quality candidate through information cascades (Deltas, Herrera, and Polborn, 2016), or, to a limited extent, by runoffs: (Bouton, 2013), (Pons and Tricaud, 2018).

Besides that, voters are also found to coordinate on past rankings or electoral results (Granzier, Pons, and Tricaud, 2023), exit polls (Morton, Muller, Page, and Torgler, 2015) or focal points such as ballot position (Forsythe and Myerson, 1991), (Marcinkiewicz, 2014). In our model a high-vote power area is a focal point itself; the paper studies the resulting voters' coordinated strategic mobility. Voters flow between districts to maximize their marginal impact on the outcome and support favored parties in destinations only if the expected risk of seats loss in the origin is not too high. Information about one's own cost of voting and the district's characteristics will suffice to determine individual decision, given aggregate flows.

Malapportionment affects voters' pivotality, shifting individual decision-making concerning both the mobility and local voting. Since voters receive positive payoffs from a supported candidate's victory only if she obtains sufficiently many votes, a rational

voter considers the maximization problem solved by other players, jointly entailing equilibrium turnout rates as in (Palfrey and Rosenthal, 1985). For unknown size of electorates, a basis for many modern applications has been Poisson games, introduced in (Myerson, 1998), (Myerson, 2000), where each electorate is determined by a random draw from the Poisson distribution, and recently used in (Smith, Bueno de Mesquita, and LaGatta, 2017) and (Ekmekci and Lauermann, 2022). Each agent, while deliberating whether to vote or abstain, needs to take this uncertainty into consideration. Poisson games, serving as a micro-foundation of our model, provide a convenient tool to model elections in large populations without arbitrarily dividing same-preference voters into different behavioral types. The original framework admits only low-turnout equilibria, but high turnout rates can be retrieved if ethical motives (as in (Jankowski, 2007) and (Feddersen and Sandroni, 2006)) or opinion polls (Diermeier and Van Mieghem, 2008) are considered.

To face a related challenge of matching observed turnouts with theoretical predictions, empirically oriented works have recently followed the choice models standardly applied in empirical industrial organization, such as (Berry, Levinsohn, and Pakes, 1993). Kawai, Toyama, and Watanabe (2021) as well as Berry, Cox, and Haile (2025) relax the assumption that voters compute their true probability of breaking the tie and focus on the perceived pivotality instead. As a result, both the benefit and the cost function include all sorts of subjectively valued factors and can be conveniently estimated from the data. This paper follows these frameworks by focusing on the subjectively perceived pivotalities as a function of *true* pivotality. Notably, individual choice translates in aggregation to turnout rates, often characterized by spatial spillovers. A complementary strand in empirical literature addresses this problem by focusing on spatial spillovers in turnout modeling: (Hoffer, 2014), (Lacombe, Holloway, and Shaughnessy, 2014), (Keele, Titiunik, and Zubizarreta, 2015). Our paper provides a micro-founded mechanism of voters' behavioral response to cross-district flows. High-turnout areas may "export" voters into nearby constituencies, with local voters mobilizing in response.

The setting examined in this paper concerns the reverse dynamic to most of the cases above. Instead of political actors changing district boundaries, seat allocations, or registration rules to shape electoral incentives, voters themselves respond to unequal vote power by relocating their ballots across fixed constituencies. In this sense, strategic mobility is a voter-driven counterpart to institutional distortions in representation.

3 Theoretical model

We start by developing a theoretical framework for the analysis of an interaction between mobilization, perceived marginal pivotality and heterogeneous costs of casting a ballot. Formalizing the model aims at building an understanding of the mechanics behind voters' electoral decisions. This helps to identify the strategic channel and measure the scale of the strategic mobility in the first-stage of the empirical part. Additionally, the theoretical model provides ex ante predictions about which factors affect the mobilization and what equilibrium effects are expected conditional on the realized flow of voters.

The theoretical framework establishes the existence of a voters flow that satisfies individual rationality and provides aggregation consistent with expectations. In such case, the mass of voters choosing to travel given the anticipated electoral environment induces exactly the expected electoral conditions. If multiple potential equilibrium voters flows exist, we select the one resulting in the lowest aggregate strategic mobility. The rest of this section is organized as follows. First, the theoretical framework is set up. Then, we describe individual choice of voters and their aggregation. Subsection 3.4 defines the equilibrium concept. Then, the equilibrium flow and resulting aggregated electoral environment are characterized. Finally, section 3.6 states a set of predictions and design implications of the theoretical framework, which are then used to structure the empirical model and test its outcomes.

3.1 Setup

We consider an environment with two voting locations $j \in \{A, B\}$ and two political parties $p \in \{L, R\}$. There are N_j voters initially assigned to each constituency j. A fraction γ_j of these voters supports party L, while the remaining fraction supports party R. Let

$$\pi_{jL} = N_j\gamma_j, \qquad \pi_{jR} = N_j(1-\gamma_j)$$

denote the numbers of voters initially assigned to constituency j who support parties L and R, respectively. Each voter may abstain, vote in the assigned constituency, or re-register and vote in the other constituency.

Let X_{jp} denote the number of ballots cast for party p in constituency j, and let

$$X_j = X_{jL} + X_{jR}$$

denote the total number of ballots cast in constituency j. We use $V(\cdot)$ to map the number of ballots cast into vote power, which measures the relative impact of an individual vote in a given constituency:

$$v_j = V(X_j), \qquad V'(\cdot) < 0.$$

Normalizing the number of seats in each constituency to one, this mapping is given by:

$$V(X_j) = \frac{1}{X_j}.$$

Thus, an individual vote carries less weight when more ballots are cast in the constituency.

The electoral margin in constituency j is

$$m_j = |X_{jL} - X_{jR}|.$$

We use $D(\cdot)$ to map the electoral margin into closeness:

$$d_j = D(m_j), \qquad D'(\cdot) < 0.$$

Hence, a smaller absolute difference between the numbers of votes cast for the two parties corresponds to a closer electoral race and therefore to a higher value of d_j. The party vote totals X_{jp}, total ballots X_j, and electoral margin m_j are the result of aggregating individual decisions of voters given the anticipated electoral conditions.

Define the anticipated electoral environment across the two constituencies as:

$$\omega = (v_A, d_A, v_B, d_B) \in \Omega, \quad \Omega \subseteq \mathbb{R}^4_{++}$$

where Ω denotes the set of feasible electoral environments. It consists of common expectations about the vote power and electoral closeness in both constituencies. Vote power and closeness of race in each location aggregate individual voting decisions. Given rational expectations, an individual, atomistic voter takes them as given when deciding where to vote.

Given a common expectation regarding the voting environment ω, the perceived marginal pivotality of voting in constituency j is summarized by

$$q_j = q(v_j, d_j).$$

This function is motivated by a Poisson voting game in which the marginal probability that an individual vote affects the electoral outcome depends on the institutional weight of the vote and on the expected margin between the parties. We do not solve the underlying Poisson voting game explicitly, since the purpose of the model is to study the location choice induced by differences in marginal pivotality. Instead, we assume that marginal pivotality is increasing in both vote power and closeness:

$$q_v(v_j, d_j) > 0, \qquad q_d(v_j, d_j) > 0.$$

Thus, a ballot has a larger perceived marginal impact when it is cast in a constituency with higher vote power or in a closer electoral race[5].

Let $P > 0$ denote the payoff associated with a voter's preferred electoral outcome. A ballot affects this payoff through its marginal effect on the probability of that outcome, captured by $q(v_j, d_j)$. The marginal expected benefit from voting in constituency j is therefore

$$Pq(v_j, d_j),$$

and for ease of exposition we normalize P to one.

The set of available actions is

$$\mathcal{A} = \{0, H, T\},$$

where 0 denotes abstention, H denotes voting in the home constituency, and T denotes re-registering and traveling to vote in the other constituency.

Voters differ in their private cost of participating in the election. Each voter i draws a cost

$$c_i \sim F(c).$$

We assume that F is continuous on $\mathbb{R}_+$, with $F(0) = 0$, and has density $f(c) > 0$ on its support. We assume that F is identical across constituencies and party types. As the party preference does not directly enter the location-choice utility (it is symmetric for both parties), supporters of different parties assigned to the same constituency face an identical problem.

[5]Poisson voting game provides a natural microfoundation for the assumptions on the direction pivotality function change. In an electoral game a voter is pivotal when the vote is tied or won by a single-vote margin. With votes per party determined by two independent Poisson distributions, the probability of such event decreases – or the pivotality declines – as the electorate becomes larger or the margin between parties grows. See Myerson (1998, 2000).

The cost c_i captures the time, effort, and financial cost of voting in the assigned constituency. Traveling away from the initial – or home – constituency increases the cost of participation. We assume that travel scales the voting cost by a factor

$$\alpha > 1.$$

Thus, voting in the home constituency costs c_i, while re-registering and voting in the other constituency costs αc_i. Abstaining bears no cost.

A voter is characterized by a type

$$t_i = (j_i, p_i, c_i) \in \mathcal{T},$$

where

$$\mathcal{T} = \{A, B\} \times \{L, R\} \times \mathbb{R}_+$$

and the component j_i denotes the constituency to which voter i is initially assigned, p_i denotes the party she supports, and c_i denotes her private voting cost.

3.2 Individual choice

Given a common expectation regarding the electoral environment $\omega = (v_A, d_A, v_B, d_B)$, an individual voter takes the marginal pivotality terms $q(v_A, d_A)$ and $q(v_B, d_B)$ as given. Consider a voter of type

$$t_i = (j_i, p_i, c_i).$$

The party preference p_i determines which party receives the vote, conditional on participation. When choosing a specific voting location, however, the voter compares the marginal pivotality of voting in each constituency with the associated cost.

The utility from abstaining is normalized to zero:

$$U_i(0; \omega, t_i) = 0.$$

The utility from voting in the home constituency is

$$U_i(H; \omega, t_i) = q(v_{j_i}, d_{j_i}) - c_i.$$

The utility from re-registering and voting away is

$$U_i(T; \omega, t_i) = q(v_{-j_i}, d_{-j_i}) - \alpha c_i.$$

A pure strategy is a measurable mapping from the expected electoral environment and the voter type into an action:

$$s : \Omega \times \mathcal{T} \to \mathcal{A}.$$

Given ω, voter i chooses an action that solves

$$s(\omega, t_i) \in \arg\max_{a \in \mathcal{A}} \{0,\ q(v_{j_i}, d_{j_i}) - c_i,\ q(v_{-j_i}, d_{-j_i}) - \alpha c_i\}.$$

This defines the individual best-response problem conditional on the common, anticipated electoral environment.

3.3 Aggregation

Given an anticipated electoral environment ω and a strategy s, individual choices aggregate into re-registration flows, party vote totals, and the effective electorate.

The number of voters of party type p assigned to constituency j who vote in their home constituency is

$$H_{jp}(s;\omega) = \pi_{jp} \int \mathbb{1}\{s(\omega, (j,p,c)) = H\}\, dF(c).$$

The number of voters of party type p assigned to constituency j who re-register and vote in the other constituency is

$$M_{(j,-j),p}(s;\omega) = \pi_{jp} \int \mathbb{1}\{s(\omega, (j,p,c)) = T\}\, dF(c).$$

Thus, $M_{(j,-j),p}(s;\omega)$ denotes the flow of party-p voters from constituency j to constituency $-j$.

Aggregating over party types, the total flow from constituency j to constituency $-j$ is

$$M_{(j,-j)}(s;\omega) = \sum_{p \in \{L,R\}} M_{(j,-j),p}(s;\omega).$$

We define the net inflow into constituency j as

$$\Delta M_j(s;\omega) = M_{(-j,j)}(s;\omega) - M_{(j,-j)}(s;\omega).$$

A positive value of $\Delta M_j(s;\omega)$ means that constituency j is a net destination of re-registering voters, while a negative value means that it is a net origin. Given the two-location environment:

$$\Delta M_j(s;\omega) = -\Delta M_{-j}(s;\omega).$$

The effective electorate in constituency j after re-registration is

$$\widetilde{N}_j(s;\omega) = N_j + \Delta M_j(s;\omega).$$

Thus, the effective electorate increases when more voters flow into constituency j than flow out of it, and decreases when constituency j is a net origin of re-registration.

The number of votes cast for party p in constituency j is

$$X_{jp}(s;\omega) = H_{jp}(s;\omega) + M_{(-j,j),p}(s;\omega).$$

The first term captures voters assigned to constituency j who vote at home. The second term captures voters of party type p who are assigned to the other constituency but re-register and vote in constituency j.

Total votes cast in constituency j are

$$X_j(s;\omega) = X_{jL}(s;\omega) + X_{jR}(s;\omega).$$

The turnout rate in constituency j is

$$r_j(s;\omega) = \frac{X_j(s;\omega)}{\widetilde{N}_j(s;\omega)}.$$

Finally, the party margin in constituency j is

$$m_j(s;\omega) = \left|X_{jL}(s;\omega) - X_{jR}(s;\omega)\right|.$$

The aggregate objects

$$\left(\Delta M_j(s;\omega), \quad \widetilde{N}_j(s;\omega), \quad X_{jp}(s;\omega), \quad m_j(s;\omega)\right)$$

jointly determine the realized aggregate electoral environment. Net re-registration flows determine the effective electorate and therefore the turnout denominator. Party vote totals determine both the number of ballots cast, which enters vote power, and the electoral margin, which enters closeness. In particular, the equilibrium realization of the aggregate objects must be consistent with anticipated electoral environment ω.

3.4 Equilibrium

We define equilibrium as a static Bayesian equilibrium with rational expectations over aggregate electoral conditions. Voters differ in their private cost draw and choose simultaneously after observing their type. Individuals share a common expectation regarding the equilibrium electoral environment

$$\omega^* = (v_A^*, d_A^*, v_B^*, d_B^*).$$

An equilibrium consists of a strategy s^* and an anticipated electoral environment ω^* such that:

1. individual strategies are optimal given ω^*;
2. the anticipated ω^* is correct: the realized vote power and closeness of race across constituencies are equal to the rationally expected values.

Voters choose optimally given the expected equilibrium electoral conditions. For every voter type $t = (j, p, c)$, the equilibrium strategy satisfies

$$s^*(\omega^*, t) \in \arg\max_{a \in \mathcal{A}} \left\{0,\ q(v_j^*, d_j^*) - c,\ q(v_{-j}^*, d_{-j}^*) - \alpha c\right\}.$$

Using the aggregate objects defined above and evaluating them at (s^*, ω^*), equilibrium vote power satisfies

$$v_j^* = V\left(X_j(s^*; \omega^*)\right),$$

and equilibrium closeness satisfies

$$d_j^* = D\left(m_j(s^*; \omega^*)\right),$$

for each $j \in \{A, B\}$. Thus, ω^* equals the anticipated electoral environment and the realized equilibrium electoral conditions given individual strategies on-path.

Finally, equilibrium marginal pivotality in constituency j is summarized by:

$$q_j^* = q(v_j^*, d_j^*).$$

3.5 Equilibrium characterization

First, consider a benchmark electoral game, in which re-registration is not permitted. Voters can choose between voting in their assigned constituency or abstaining. Let ω^{NT} denote the rationally expected electoral environment aggregating voters' decisions when the travel to vote is not allowed. Similarly, denote by s^{NT} individual strategies in such restricted environment. Together, a pair (ω^{NT}, s^{NT}) constitutes a rational-expectations equilibrium in the benchmark no-travel game. Then, consider a case in which:

$$q(v_A^{NT}, d_A^{NT}) < q(v_B^{NT}, d_B^{NT}),$$

i.e. the anticipated no-travel benchmark pivotality is higher in constituency B than in constituency A. This difference may result from vote power, closeness of the race, or from both. Having described this no-travel benchmark, let us now consider the direction of flows in the regular environment.

Lemma 1 (Direction of flows)**.** *Suppose that an electoral environment ω satisfies:*

$$q_B(\omega) > q_A(\omega),$$

where q_i denotes a pivotality in constituency $i \in \{A, B\}$. Then no voter assigned to constituency B strictly prefers to re-register and vote in constituency A. Moreover, a positive mass of voters assigned to A can re-register and vote in B only if $q_B(\omega) > q_A(\omega)$:

$$M_{(A,B)}(s, \omega) > 0 \implies q_B(\omega) > q_A(\omega).$$

Proof. See Appendix A.

Conditional on the no-travel benchmark ordering $q_B^{NT} > q_A^{NT}$, we focus the case in which the relevant strategic flow is from A to B.:

$$M \equiv M_{(A,B)}, \qquad M_{(B,A)} = 0,$$

so that the net flows are:

$$\Delta M_A = -M, \qquad \Delta M_B = M.$$

For a candidate one-way flow M, let

$$\omega(M) = (v_A(M), d_A(M), v_B(M), d_B(M))$$

denote the aggregate electoral environment consistent with mappings V and D, conditional on that candidate flow. Once we restrict attention to one-way flows from A to B, the aggregate flow M summarizes the effect of re-registration on the anticipated electoral environment. Let

$$q_j(M) \equiv q_j(\omega(M)) = q(v_j(M), d_j(M)).$$

Define

$$\Delta q(M) = q_B(M) - q_A(M).$$

Lemma 1 implies that strategic re-registration can be positive in at most one direction. Any positive flow from A to B requires:

$$q_B(M) > q_A(M).$$

If the two constituencies had equal pivotality, traveling to B would provide no additional marginal impact relative to voting at home, while requiring the higher cost αc_i. Hence no voter with $c_i \geq 0$ would choose to travel[6].

The individual decision of a voter assigned to constituency A is defined by the three cutoffs. Let

$$c_{TH}(M) = \frac{\Delta q(M)}{\alpha - 1}$$

denote the cost at which a voter is indifferent between traveling to B and voting at home. Let

$$c_{T0}(M) = \frac{q_B(M)}{\alpha}$$

denote the cost at which traveling to B is indifferent to abstention, and let

$$c_{H0}(M) = q_A(M)$$

denote the cost at which voting at home is indifferent to abstention.

Proposition 1 (Cutoff structure)**.** *Consider a positive-flow M from A to B, i.e., $\Delta q(M) > 0$. The best response action of a voter assigned to constituency A given*

[6]We assume that when a voter is indifferent between traveling and another action, she does not travel. If voting at home is weakly optimal, she votes at home; otherwise she abstains.

the flow M is characterized by the following cutoff rule:

$$a_i^{BR}(A, c_i; M) = \begin{cases} T, & \text{if } c_i < \bar{c}_T(M), \\ H, & \text{if } c_{TH}(M) \leq c_i \leq c_{H0}(M), \\ 0, & \text{otherwise}, \end{cases}$$

where

$$\bar{c}_T(M) = \min\{c_{TH}(M), c_{T0}(M)\}.$$

Voters assigned to constituency B do not travel to A. Their best response action is described by:

$$a_i^{BR}(B, c_i; M) = \begin{cases} H, & \text{if } c_i \leq q_B(M), \\ 0, & \text{otherwise.} \end{cases}$$

Proof. See Appendix A.

Proposition 1 gives the characterization of an individual decision depending on the relative perceived marginal pivotality between constituencies and individual cost. Strategic mobility is determined by the distribution of individual costs, and a travel cutoff $\bar{c}_T(M)$ – which in turn depends on the marginal impact of an individual vote in each location given the flow of voters. In the empirical setting, the main case of interest is the comparison between voting at home and voting in a more pivotal destination. The travel-abstention constraint is retained in the model for completeness, since sufficiently high destination pivotality may also induce participation among voters who would otherwise abstain. Local turnout among voters assigned to A is determined by the interval between $c_{TH}(M)$ and $c_{H0}(M)$, while local turnout among voters assigned to B is determined by $q_B(M)$.

The cutoff rule identifies the share of voters whose costs make traveling optimal. When $\Delta q(M) > 0$, voters assigned to A travel to B if their cost is below $\bar{c}_T(M)$. Hence the mass of voters who would choose to travel given the expected electoral environment associated with M is:

$$N_A F(\bar{c}_T(M)).$$

If $\Delta q(M) \leq 0$, Lemma 1 implies that traveling to B is not optimal for any positive mass of voters assigned to A. We therefore define the aggregate flow response as

$$\Phi(M) = \begin{cases} N_A F(\bar{c}_T(M)), & \text{if } \Delta q(M) > 0, \\ 0, & \text{if } \Delta q(M) \leq 0. \end{cases}$$

Because the cutoff rule is the same for supporters of both parties within a constituency, the party-specific components of the flow preserve the party composition of the origin. For any M such that $\Delta q(M) > 0$:

$$M_{(A,B),L}(M) = \gamma_A M, \qquad M_{(A,B),R}(M) = (1 - \gamma_A)M.$$

Home voting in the origin and destination follows from Proposition 1. Under the positive-flow M,

$$H_{Ap}(M) = \pi_{Ap} \left[F(c_{H0}(M)) - F(c_{TH}(M)) \right]_+ ,$$

and

$$H_{Bp}(M) = \pi_{Bp} F(q_B(M)).$$

Party-specific flows and home voting enter the aggregation identities characterizing the electoral environment $\omega(M)$.

Proposition 2 (Equilibrium characterization)**.** *An equilibrium flow from A to B is a fixed point of the mapping:*

$$M^* = \Phi(M^*),$$

where Φ is the aggregate flow response defined above. The associated equilibrium expectation is $\omega^* = \omega(M^*)$.

At any equilibrium flow M^*, party-specific flows preserve the initial party composition in the origin:

$$M^*_{(A,B),L} = \gamma_A M^*,$$

and

$$M^*_{(A,B),R} = (1 - \gamma_A) M^*.$$

If F, q, $v_j(M)$, and $d_j(M)$ are continuous in M, an equilibrium flow exists. If multiple equilibrium flows exist, we select the minimal equilibrium flow

$$M^{\min} = \min \left\{ M \in [0, N_A] : M = \Phi(M) \right\}.$$

Given $M^{\min}$, the individual strategy is characterized by Proposition 1. The remaining aggregate outcomes are obtained from the aggregation identities described in Section 3.3 and the equilibrium consistency conditions in Section 3.4.

Proof. See Appendix A.

Strategic flows update pivotality in equilibrium. A flow from A to B changes the number of ballots cast and the electoral margin in both locations. Since

$$v_j(M) = V(X_j(M)) \qquad \text{and} \qquad d_j(M) = D(m_j(M)),$$

re-registration affects both components of marginal pivotality. This feedback generates the fixed-point structure in Proposition 2: the flow changes the aggregate electoral environment, which in turn determines how many voters want to move.

Proposition 2 establishes that at least one flow satisfying individual optimality and aggregate consistency exists. Proposition 1 defines individual best responses given any flow M. Integrating these best responses over the cost distribution gives the flow response $\Phi(M)$. An equilibrium requires that the candidate flow equals the mass of voters who optimally choose to travel in the electoral environment induced by that flow. If multiple values M exist that satisfy that condition, we select the equilibrium with the smallest number of strategic re-registrations. This is a conservative selection rule that attributes to strategic mobility only the minimum amount of movement needed to satisfy individual optimality and aggregate consistency.

3.6 Comparative statics and empirical implications

The theoretical framework carries a number implications for the empirical part of the analysis. First, it provides information on factors that are likely to predict strategic re-registrations in the empirical setting (Prediction 1). The theoretical framework also indicates that higher cost associated with travel limits the flow between constituencies, which justifies the use of inverse spatial weights (Design implication 1). Furthermore, comparative statics provide testable predictions regarding the direction of changes in re-registrations and – as a consequence – in equilibrium outcomes depending on the characteristics of each constituency (Prediction 2 and 3). Finally, the model clarifies how strategic flows affect turnout incentives in equilibrium, and why observed turnout must be separated into a mechanical component and a behavioral response of local voters (Design implication 2).

Design implication 1: travel costs constrains strategic re-registration. The travel friction parameter α lowers the payoff from re-registering relative to voting

at home. In every case the travel cutoff is decreasing in α:

$$\frac{\partial}{\partial \alpha}\left(\frac{\Delta q(M)}{\alpha - 1}\right) < 0, \qquad \frac{\partial}{\partial \alpha}\left(\frac{q_B(M)}{\alpha}\right) < 0.$$

Thus, higher travel costs reduce the set of voters for whom re-registering is optimal. Our empirical analysis will introduce a multiple-location setting, so the result hints at a spatial constraint of strategic re-registration prior to elections. Voters are more likely to consider destinations that are less costly to reach, which may correspond to locations closer to home.

This implication motivates the use of SLX-type specification with inverse spatial weights in the estimation of strategic re-registrations in the first-stage of empirical analysis. Perceived pivotality of nearby alternatives matters, but only to the extend that switching constituency is not too costly considering a potential gain.

Prediction 1: strategic re-registration responds to relative perceived pivotality. Strategic re-registration is driven by differences in perceived marginal pivotality across locations. An increase in destination pivotality raises the value of voting in B relative to voting in A. Since

$$q_v(v_j, d_j) > 0 \qquad \text{and} \qquad q_d(v_j, d_j) > 0,$$

higher vote power or greater closeness in the destination increases $q_B(M)$, raises the travel cutoff, and increases the mass of voters who find it optimal to re-register.

Conversely, higher pivotality in the origin increases the value of voting at home. When the home-vs-travel margin is relevant, an increase in $q_A(M)$ lowers

$$\Delta q(M) = q_B(M) - q_A(M),$$

reduces the travel cutoff, and decreases strategic outflows from A. Thus, the model predicts that strategic flows should originate in locations with low perceived pivotality and be directed toward locations where the marginal impact of a vote is higher.

In the empirical setting, vote power and closeness are the observable counterparts of the two components of $q(v, d)$. Closeness measures the seat-changing margin of votes, approximated by the minimum number of additional votes needed to change an allocation of marginal seat. Vote power captures a relative influence of a single ballot, i.e. how many seats are elected in a constituency by how many voters. The

empirical model therefore examines whether vote power and closeness predict strategic re-registration in 2023 elections in the directions implied by the model, using electoral results in 2019 as the closest available approximation of ω^{NT}.

Design implication 2: observed turnout change combines direct and behavioral components. Strategic re-registration may affect the realized turnout in two distinct manners. First, there is a direct effect of incoming voters in the destination, who increase the number of ballots cast in the location. The opposite direct effect results in a decrease in turnout in origins. In the language of the theoretical model, net flows ΔM_j affect the effective electorate $\widetilde{N}_j$ and change the total votes cast in a constituency X_j. Additionally, the change in the number of voters within a constituency may have an indirect, behavioral impact on non-movers in either location, which in the model is measured by a change in the number of home voters H_j.

This distinction motivates the second-stage empirical design. To study whether strategic flows changed local participation incentives, the empirical analysis must separate the direct effect of movers from the equilibrium response of local voters. In the empirical analysis we use an adjusted turnout measure – an estimated number of re-registered voters is deducted from the destination turnout calculations – and interpret the second stage as evidence on local equilibrium responses rather than as a direct effect of incoming voters raising the number of ballots cast.

Prediction 2: outflows can affect home turnout in origins. The model implies that strategic outflows can affect the participation incentives of voters who remain in the origin. For voters assigned to A, home voting is determined by the interval

$$c_{TH}(M) \leq c_i \leq c_{H0}(M),$$

where

$$c_{H0}(M) = q_A(M)$$

is the participation cutoff for voting at home. If outflows increase marginal pivotality in the origin, then $q_A(M)$ rises and the upper cutoff $c_{H0}(M)$ increases. This expands the set of voters for whom voting at home dominates abstention.

When voters leave their original constituency, they no longer cast ballots there. This reduces the number of ballots cast in A, and the vote power in the origin increases as $v_A(M) = V(X_A(M))$ and $V'(\cdot) < 0$. Higher vote power raises marginal pivotality and weakly increases the incentive for remaining voters to vote at home.

Since the cost distribution is identical across party types and party preference does not enter the location-choice utility, outflows from a given origin preserve the initial party composition in that constituency. However, the absolute margin may be reduced by flows as fewer voters cast their ballots. In this case, the equilibrium channel is the change in local marginal pivotality of voters who stay in a constituency, affected by a shift in vote power and possibly through a change of absolute margin.

This prediction maps to the second-stage analysis of turnout effects in the origin locations. If strategic outflows raise the marginal value of a home vote for remaining voters, adjusted turnout in origins may increase.

Prediction 3: inflows have an ambiguous effect on local turnout in destinations. The effect of inflows on local turnout in destinations is ambiguous. For voters assigned to B, home voting is described by:

$$H_{Bp}(M) = \pi_{Bp} F(q_B(M)).$$

Thus, local turnout in the destination increases if inflows raise $q_B(M)$, and decreases if inflows lower $q_B(M)$.

There are two channels that influence perceived pivotality in the destination. First, incoming voters increase the number of ballots cast in the destination. Since vote power is decreasing in the number of ballots cast,

$$V'(\cdot) < 0,$$

this reduces marginal pivotality via its vote power component – the vote power dilution discourages participation by local voters.

Second, inflows can change the closeness of race. If incoming voters strengthen the weaker side in the destination, the vote margin falls, closeness rises, and marginal pivotality may increase. On the contrary, if incoming voters strengthen the locally leading side, the vote margin grows, closeness falls, and marginal pivotality is expected to decrease. The sign of the local turnout response therefore depends on the direction of changes in closeness, and whether it dominates the vote power dilution channel.

In the empirical setting, strategic movers from lower pivotality urban constituencies are likely to be politically selected. As the incumbent party dominated in rural

areas, while voters from large urban areas disproportionately supported the opposition, inflows can be expected to affect the destination race closeness via increased competition in destinations.

This prediction is central to the second-stage empirical analysis of destinations. A positive effect on adjusted turnout in high-inflow destinations would suggest that the local equilibrium response due to race becoming more competitive dominates the vote power dilution channel. Conversely, weak or negative adjusted-turnout effects would be consistent with dilution dominating, or with inflows failing to improve local competitiveness.

4 Empirical analysis

The theoretical model provides insights concerning strategic flows of voters dictated by ex ante pivotality and the equilibrium reaction of local voters to these flows. Our empirical analysis follows closely these predictions and consists of two stages. In the first, we identify the size of strategic flows in the 2023 election in Poland. The exercise disentangles strategic motives for relocation (vote power and race closeness) from other channels of voters' mobility, such as regular tourism or nominal registrations in the place of actual (but otherwise unregistered) residence. We use spatial regression techniques to account for spatial substitution patterns predicted by the theory and run a counterfactual exercise by comparing predicted flows under the true scenario with the one where strategic channels have been muted.

In the second stage, the derived flows are used to explain equilibrium turnout response to voters' strategic relocation between places. We check whether locations with higher strategic inflows or outflows increase the turnout rates among non-moving voters in a faster pace than in unaffected, otherwise similar municipalities.

4.1 Data

Our dataset has been compiled using data from the Polish Electoral Office, Central Voters Register, and the Central Statistical Office (GUS). Data have been aggregated by municipality (gmina level), yielding $n = 2479$ observations in total.

With a few exceptions[7], each variable is defined for 2023 or, where necessary, for 2019 (previous parliamentary elections) to avoid endogeneity. Our key variables include: share of voters temporarily included in a municipality's registry (*Add23*), continuous index of the weight carried by a single vote within a constituency (VP)[8], as well as the binary index determining closeness of a race in a constituency (*Close*) between the incumbent Law and Justice and the democratic opposition (Civic Platform, Third Way, United Left). *Close* assumes value 1 for the top quarter of competitive races as of the 2019 election (25% with the least votes to flip one seat in a district according to the D'Hondt method), and 0 otherwise. These variables correspond to the two components of perceived marginal pivotality in the theoretical model: VP captures the leverage of an additional ballot, while $Close$ captures proximity to a seat-changing margin. Besides, we include a rich set of socio-economic (salary, urbanization, population density, unemployment, education), infrastructural (roads, bus stops, hotel beds), and demographic covariates (population, share of women, share of pensioners, unregistered citizens, age, population change).

The structure of this dataset is closely related to challenges that our empirical specification must face. Unfortunately, individuals cannot be tracked in their decisions to change voting location, and we also cannot assign voters' flows to unique origin-destination pairs. Inflows aggregated at the municipality level are as fine-grained as one can get here[9]. At the same time, our control variables are chosen to mitigate the existence of other reasons why voters might change their location, including regular tourism and outdated registrations systematically "adding" votes in big cities[10]. Finally, we take two empirical assumptions justifying the following specification:

Assumption 1. *Strategic mobility was absent in the previous parliamentary election of 2019.*

[7]Share of population with a university degree is available only for 2021 and at the county-level (powiat), and median age only at the county-level.

[8]Ratio of the number of seats elected in a district to the number of voters in the previous election (2019). The variable is standardized to 1, such that values below it denote underrepresented districts and above – the overrepresented.

[9]Although the size of 2023 outflows per municipality is also available, we do not employ it in our main specifications. For previous elections, these data have not been centrally collected, preventing a comparison with past trends. There is, however, serious reason to expect high persistence in voters' mobility (as also further confirmed in our data), so for safety reasons, we decided to drop this variable.

[10]Fortunately, there is no need to include data on weather in the two compared elections. Both in 2019 and 2023, it took place on a relatively warm and sunny October Sunday, 13.10 and 15.10 respectively.

The assumption implies that the observed 2019 variation is the baseline (natural) mobility. We will therefore focus on the measurement of 2023 deviations from this trend. To provide further evidence, Figure 2 illustrates the change in share of temporary voters between the two elections across municipalities. Notably, the big mass of distribution moves to the right between the elections[11].

Assumption 2. *Costs of switching location grow in distance.*

The second assumption is motivated by the theoretical model which indicates that the willingness to travel for voting decreases in the distance between locations. From this, it follows that voters' mobility is characterized by spatial frictions. As a result, the problem can be framed as one of local spillovers.

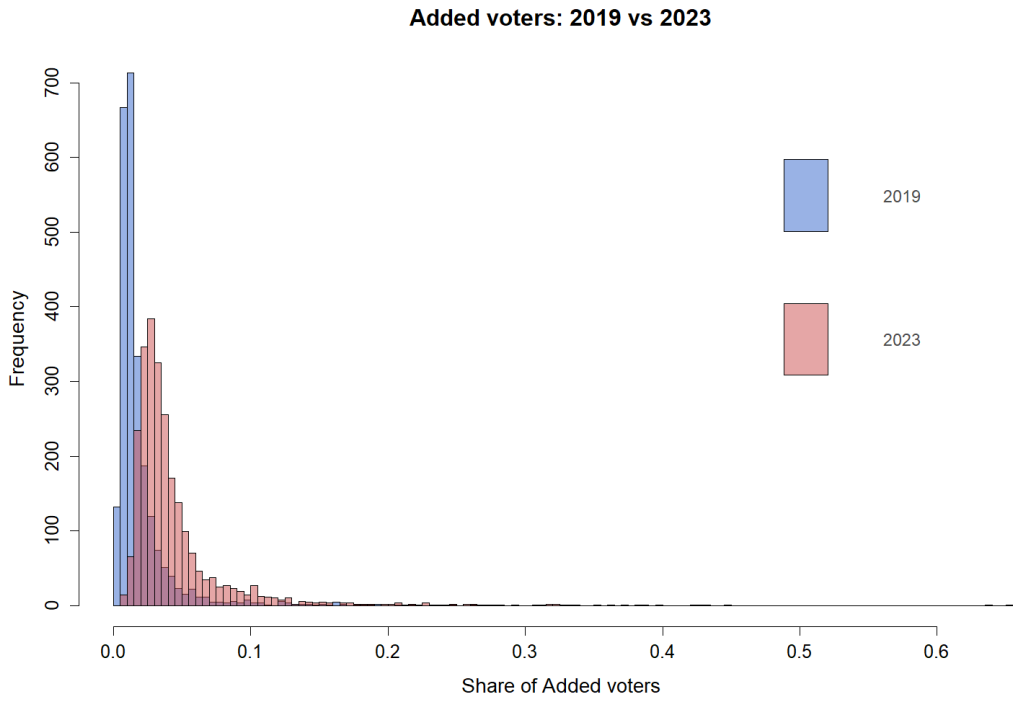


Figure 2: Share of new voters temporarily included in a registry, by municipality

[11]Outliers with surprisingly high shares of outside voters (reaching in most extreme cases even 60% of total votes in a municipality) can be easily explained as regular tourists inflows and accounted for by our control variables.

4.2 First stage: modeling strategic flows

Since a voter's decision to vote home or relocate depends not only on the characteristics of his origin municipality but also on the available alternatives, our theory predicts significant spatial patterns. Two otherwise similar locations can observe diverse inflows of voters if they will be surrounded by sufficiently different neighbors. Therefore, to correctly account for these spatial substitution patterns, our empirical model includes spatial lags of explanatory variables (SLX model) to predict the size of total (strategic and non-strategic) inflows into each municipality. In other words, a unit's inflows of voters will be explained both by its own characteristics as well as its neighbors', within a 30-kilometer range. Consider the following specification for a municipality j belonging to a constituency d:

$$\begin{aligned} Add23_j = \beta_0 + \beta_1 VP_d + \beta_2 Close_d + \beta_3 (VP_d \times Close_d) + \mathbf{Z_j}\boldsymbol{\gamma} \\ + \mathbf{w}_j'\mathbf{VP}\,\theta_1 + \mathbf{w}_j'\mathbf{Close}\,\theta_2 + \mathbf{w}_j'(\mathbf{VP} \circ \mathbf{Close})\,\theta_3 + \mathbf{w}_j'\mathbf{Z}\boldsymbol{\delta} + \epsilon_j. \end{aligned} \tag{1}$$

The dependent variable, $Add23_j$ is a 2023 share of incoming voters in municipality j, relative to the total number of voters who turned out in j. Variables VP_d and $Close_d$ are a unit's own vote power and a race closeness defined at a constituency level (such that different municipalities in the same constituency are assigned the same value), and $\mathbf{Z_j}$ is a vector of its further characteristics. $\mathbf{VP}, \mathbf{Close}$ and $\mathbf{Z}$ (without subscript) are n-length vectors, accounting for relationships between different units. Coefficients $(\beta_1, \beta_2, \beta_3)$ and $(\theta_1, \theta_2, \theta_3)$ then identify, respectively, direct relationships between the explanatory and the explained variables, and the strength of spatial spillovers. Term $\mathbf{w}_j'$ represents the j-th row of a row-normalized matrix $\mathbf{W} = [w_{jk}]$, consisting of inversed-distance spatial weights assigned to connections between municipalities j and k, according to the formula:

$$w_{jk} = \frac{\tilde{w}_{jk}}{\sum_j \tilde{w}_{jk}} \quad \text{and} \quad \tilde{w}_{jk} = \begin{cases} \frac{1}{\text{dist}(j,k)} & \text{if } j \neq k \text{ and } \text{dist}(j,k) < D \\ 0 & \text{otherwise,} \end{cases} \tag{2}$$

where $\text{dist}(j,k)$ is the Euclidean distance between the centroids of j and k, and $D = 30\text{km}$.

The model in (1) thus attempts to explain variation in incoming voters' shares,

$Add23_j$, focusing on the vote power and closeness of the electoral race in a constituency d where j is located. By including spatial relationships, we also allow the characteristics of nearby units to contribute to its value. If voters consider switching their vote location based on a rational premise, they compare their expected marginal impact on the election between different places. By construction, a municipality surrounded by neighbors in the same d will observe no spatial effects, as the value of VP and $Close$ will be the same across the units. Variation in values is, however, expected to occur closer to a constituency border. The choice of inverse-distance weights in $\mathbf{W}$ secures that more distant units will contribute less to each other's dependent variable. This relates to the theoretical notion that travel is costly and increase in a distance lowers the incentive to re-register. At the same time, with the threshold of 30 kilometers, each municipality has a positive number of neighbors (between 3 and 51, with a mean number of links at 22.39). Consequently, the threshold provides enough variation in the number of links without risking the local scale of the phenomenon. In the Appendix, Table B.1 considers alternative distance thresholds.

The regression results presented in Table 1 below test the main flow prediction of the theoretical model: strategic re-registration should respond positively to a municipality's own pivotality and negatively to the pivotality of nearby alternatives. To account for pre-existing voters' mobility, $\mathbf{Z}$ also contains $Add19$, a variable specifying shares of incoming voters in the previous 2019 parliamentary election. Besides, we also include a binary border indicator, assuming 1 if a municipality lies at a constituency border. The table begins with a crudest version of the model (no race closeness or control variables) presented in the first column, with further elements added progressively. The third column presents the model from equation (1) without the interaction terms (β_3 and $\theta_3 = 0$), while the last one considers its full version. The number of observations used in the regression slightly falls compared to our full sample due to missings.

For the sake of simplicity, let us dismiss interactions for a moment. Not surprisingly, voters' mobility is characterized by high persistence across time (high coefficients at $Add19$). However, in line with our theoretical predictions, voters also tend to self-allocate themselves along the vote power. The higher VP in j, the bigger relative influx of voters it observes, and, at the same time, the higher value VP takes in j's close neighbors, the lower this influx in j will be. In other words, the voters seem to choose their location of voting rationally, given vote power in nearby municipalities, robustly to municipality characteristics. Slightly less clear patterns are observed for

Table 1: Stage 1 results

	Dependent variable: Add23			
	Vote	(1)+Close	(2)+Controls	(3)+Interaction
	(1)	(2)	(3)	(4)
Add19	1.032***	1.031***	0.886***	0.879***
	(0.017)	(0.017)	(0.017)	(0.017)
VP	0.070***	0.069***	0.077***	0.015
	(0.012)	(0.012)	(0.011)	(0.012)
Border	0.000	0.000	0.001	0.001
	(0.001)	(0.001)	(0.001)	(0.001)
Close		−0.004	−0.004	−0.241***
		(0.003)	(0.002)	(0.024)
Close × VP				0.231***
				(0.023)
lag_Add19	0.273***	0.264***	0.051	0.037
	(0.035)	(0.035)	(0.041)	(0.040)
lag_VP	−0.142***	−0.139***	−0.075***	0.003
	(0.015)	(0.015)	(0.015)	(0.016)
lag_Border	−0.001	−0.001	−0.003	−0.004
	(0.003)	(0.003)	(0.002)	(0.002)
lag_Close		0.007*	0.007*	0.357***
		(0.003)	(0.003)	(0.030)
lag(Close × VP)				−0.341***
				(0.029)
Controls	no	no	yes	yes
Observations	2429	2429	2429	2429
Note:				*p<0.05; **p<0.01; ***p<0.001

the closeness of race (positive coefficient in the spatial lag) and for the border indicator, which turns out statistically insignificant.

When the interaction between vote power and race closeness is added, the results seem to reverse at first glance. Vote power is now statistically significant only in the interaction term, and the coefficient of $Close$ is negative. However, one can easily see that for high-competition municipalities ($Close = 1$), the effect of vote power will be the same as before and null otherwise, strengthening the previous finding of voters' strategic self-allocation. To fully assess the correlation of $Close$ with the dependent variable, we compute its marginal effect across the distribution of vote power and illustrate it in Figure 3.

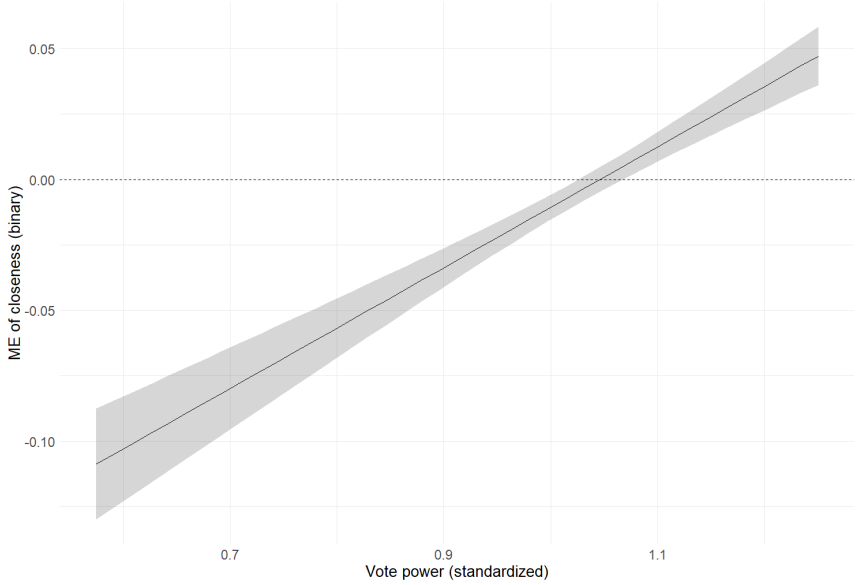


Figure 3: Marginal effect of race closeness

Consistently with our theory, the total effect of race closeness is growing in vote power. Once the latter exceeds the threshold of 1 (constituency neither underrepresented nor overrepresented given the number of seats and the number of voters), the effect becomes positive. The closer the race in the electoral district, the more additional registered voters in municipalities within it.

Therefore, the two most comprehensive versions of our empirical framework are consistent with the model's prediction that strategic re-registration responds to relative

perceived pivotality. Voters tend to self-allocate across alternative locations according to the vote power and the closeness of the electoral race, with substantial spatial components of this relationship. To evaluate the strength of this strategic channel, we conduct a counterfactual with equalized vote power and compare its results to the predicted values of (1).

Compare the fitted values given our empirically estimated model:

$$\begin{aligned}\widehat{Add23}_j = \hat{\beta}_0 + \hat{\beta}_1 VP_d + \hat{\beta}_2 Close_d + \hat{\beta}_3 (VP_d \times Close_d) + \mathbf{Z_j}\hat{\boldsymbol{\gamma}} \\ + \mathbf{w}_j' \mathbf{VP}\, \hat{\theta}_1 + \mathbf{w}_j' \mathbf{Close}\, \hat{\theta}_2 + \mathbf{w}_j' (\mathbf{VP} \circ \mathbf{Close})\, \hat{\theta}_3 + \mathbf{w}_j' \mathbf{Z}\hat{\boldsymbol{\delta}}\end{aligned} \tag{3}$$

with a counterfactual where each district is equally represented ($VP_d = 1, \forall d$):

$$\begin{aligned}\widehat{Add23}_j^{CF} = \hat{\beta}_0 + \hat{\beta}_1 \cdot 1 + \hat{\beta}_2 Close_d + \hat{\beta}_3 (1 \times Close_d) + \mathbf{Z_j}\hat{\boldsymbol{\gamma}} \\ + \mathbf{w}_j' \mathbf{1}\, \hat{\theta}_1 + \mathbf{w}_j' \mathbf{Close}\, \hat{\theta}_2 + \mathbf{w}_j' (\mathbf{1} \circ \mathbf{Close})\, \hat{\theta}_3 + \mathbf{w}_j' \mathbf{Z}\hat{\boldsymbol{\delta}}.\end{aligned} \tag{4}$$

This counterfactual shuts down the vote-power component of perceived pivotality while holding closeness, spatial structure, and municipal characteristics fixed. The difference between $\widehat{Add23}$ and $\widehat{Add23}^{CF}$ (multiplied by total count of votes) will give us the size of *strategic* voters' mobility, that is, the number of voters in each municipality that decided to switch their location due to vote power:

$$Add_diff \equiv (\widehat{Add23} - \widehat{Add23}^{CF}) \cdot (\text{total votes}).$$

Where positive, *Add_diff* will denote voters' inflows, where negative – their outflows. As stated by Lemma 1, the framework allows us to consider only unidirectional flows and discard two-way flows of voters as irrelevant. In consequence, estimated strategic mobility identifies both total and net flows. Constraining aggregated outflows to match (lower) total inflows and calibrating the biggest flows by the ratio of the estimation's precision[12], total strategic flows yield around $33,000$ voters across the country[13]. The results of this counterfactual exercise are illustrated in Figure 4

[12]If $Add23_j$ times total voters in j is greater than or equal to 100, Add_diff_j has been multiplied by $(Add23_j / \widehat{Add23}_j)$. In this way, places with the largest voters' mobility, underestimated in the empirical model, are slightly calibrated upwards.

[13]To make sense of this number, let us note that in 2019 elections, this number of votes would suffice to flip the last seat allocated according to the D'Hondt in even 17 constituencies (if, of course, properly distributed – and discarding here the party differences). In two previous parliamentary terms (2015-2019, 2019-2023), the majority was secured by only as much as 5 seats.

below. For clarity, flows between $(-200, +200)$ of individual voters have been muted in the map, with remaining outflows (inflows) depicted in red (blue). Clearly enough, origin areas are mostly big cities (Warsaw, Gdańsk, and Poznań) with some of their suburbs, while destinations are primarily represented by surrounding municipalities. Due to spatial frictions and differing vote power across constituencies, voters' movement concentrates along constituency borders, consistently with the spatial constraint implied by travel costs in the model. Notably, while the strategic mobility is most clearly seen around Warsaw in the middle of the map and in the north-western Poland, it is virtually absent in the eastern part of the country, most dominated by the Law and Justice party.

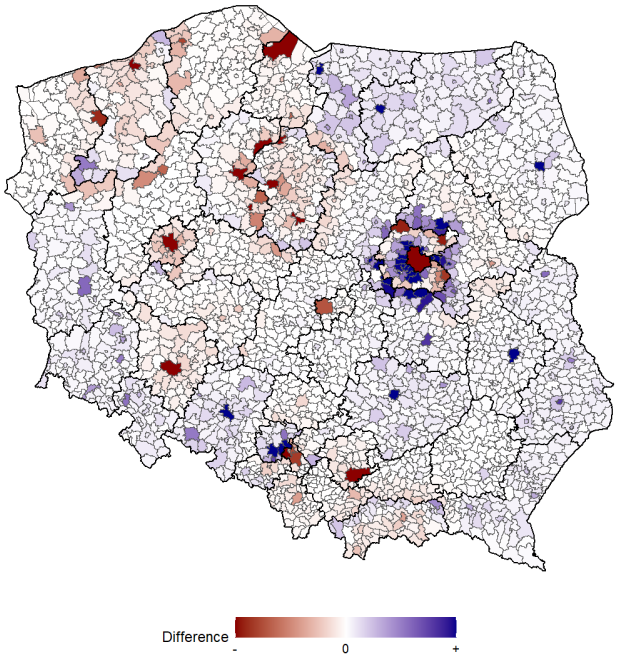


Figure 4: Spatial distribution of *Add_diff*. White values: $Add_diff \in (-200, 200)$, red $Add_diff \leq -200$ and blue $Add_diff \geq 200$

4.3 Second stage: turnout equilibrium response

Given estimated strategic flows, we evaluate equilibrium turnout responses in origins and destinations. This follows the model's distinction between the direct effects of moving voters and the behavioral responses of voters who remain in their assigned

municipality (Design implication 2). For the ease of interpretation, we first divide *Add_diff* into two distinct variables, identifying shares of inflows and outflows in the total pool of voters:

$$Inflow = \begin{cases} \frac{Add_diff}{\#\text{ actual voters in } j} & \text{if } Add_diff > 0 \\ 0 & \text{otherwise} \end{cases} \tag{5}$$

$$Outflow = \begin{cases} -\frac{Add_diff}{\#\text{ actual voters in } j} & \text{if } Add_diff < 0 \\ 0 & \text{otherwise.} \end{cases} \tag{6}$$

We separate inflows and outflows because the model assigns them different equilibrium roles: outflows affect the incentives of voters remaining in origin locations, while inflows combine a direct, mechanical addition of outside voters with a possible behavioral response of locals in destinations. As our introductory specification, we estimate the following model:

$$T_j^{adj} = \alpha_0 + \alpha_1 Inflow_j + \alpha_2 Outflow_j + \mathbf{X_j}\boldsymbol{\phi} + \zeta_j, \tag{7}$$

where T_j^{adj} is the turnout rate in municipality j in 2023, adjusted by the number of additional voters: $T_j^{adj} = \frac{\#\text{ actual voters in } j - Add_diff_j}{\#\text{ eligible voters in } j - Add_diff_j}$. The adjustment removes the estimated net direct contribution of strategic flows according to the sign of *Add_diff*: it subtracts inflows in destinations and adds back outflows in origins. Vector $\mathbf{X_j}$ contains further characteristics of the unit, including its (unadjusted) turnout rate in 2019 and number of voters per polling station as a proxy for the cost of voting[14]. Current specification thus compares affected municipalities with otherwise similar, less exposed units. As documented in Figure 4, strategic flows are predicted to occur outside of bigger cities, close to constituency boundaries.

The turnout adjustment separates the behavioral response of the local population from the direct impact of voter mobility. A positive coefficient is interpreted as evidence consistent with a positive local turnout response after removing the direct arithmetic contribution of movers. Note that the theoretical setup of a voter's maximization problem explicitly rules out cases of voters relocating to another place (and thus incurring a higher participation cost) yet finally abstaining from vote. As a result, one can assume that traveling strategic voters vote with probability 1. Finally, taking relative inflows and outflows as the independent variables guarantees that results will not be driven by larger municipalities where even big absolute flows may

[14]Hotel beds, bus stops and voters per station enter in logs.

account only for a small share of voters. Additionally, we include district-level fixed effects, to account for unobserved district-level shocks affecting both local turnout and outside voters' inflows (such as political advertising). Under the D'Hondt method applied in Poland, seat payoffs for parties are district-level, regardless of outcomes in particular municipalities within a constituency. Two municipalities in the same district are therefore similarly exposed to unobserved constituency-level campaign advertising, making district fixed effects sufficient to absorb them. In this way, treated units are compared to untreated ones within a constituency. Fixed effects provide additional robustness by excluding district-specific confounders but should be treated with some caution – as per our theoretical model, the treatment operates on the level of constitutencies. Any flows into/from a municipality j in a district d will affect the vote power and the race tightness also in all other units k in d. Even if obtaining no direct flows of voters, these units should thus respond in their turnout too.

The regression results with standard errors clustered at the district level are presented in Table 2.

Table 2: Stage 2 results

	Dependent variable: T^{adj}		
	Baseline	+ Controls	+ FE
	(1)	(2)	(3)
Inflow	0.2482**	0.1905**	0.1557**
	(0.1032)	(0.0799)	(0.0731)
Outflow	0.8504***	0.8955***	0.2622
	(0.2625)	(0.2163)	(0.2684)
Turnout19	0.8260***	0.8218***	0.8113***
	(0.0146)	(0.0227)	(0.0172)
Control variables	No	Yes	Yes
Voters per station (cost proxy)	No	Yes	Yes
Clustered SE (district)	Yes	Yes	Yes
District FE	No	No	Yes
Observations	2,429	2,429	2,429
Note:			*p<0.1; **p<0.05; ***p<0.01

In line with expectations, turnout patterns reveal strong persistence over time. At

the same time, the results show a positive and significant association of both voters' inflows and outflows with turnout rates when control variables are taken into consideration. In terms of the theoretical model, the positive inflow coefficient is consistent with a destination response in which the increased competitiveness dominates vote power dilution channel. For otherwise similar municipalities, higher relative flows to or from the unit are associated with higher adjusted turnout. Notably, the finding is robust to control variables (column 2). Subtracting the flows from the dependent variable allows us to interpret the patterns as behavioral, separate from the mechanistic change due to the simple transfer of voters from one place to another. The inflows pattern is also robust to district fixed effects (column 3).

To move beyond static correlations and account for pre-existing turnout dynamics, our main specification considers changes in turnout rates across elections. Ex ante, positive outflows-turnout link allows both for outflows increasing turnout through the growing vote power channel and for high turnout places sending out more voters. The specification asks whether the exposure to higher inflows or outflows was associated with a more rapid turnout increase than in the past, given past turnout rate and its change. Consider the following dynamic model:

$$\Delta T_j^{adj} = \alpha_0 + \alpha_1 \mathit{Inflow}_j + \alpha_2 \mathit{Outflow}_j + \alpha_3 \Delta T_{j,2019} + \mathbf{X_j}\boldsymbol{\phi} + \gamma_d + \zeta_j, \tag{8}$$

The dependent variable is now the change in turnout between the elections of 2023 and 2019: $\Delta T_j^{adj} \equiv T_j^{adj} - T_{j,2019}$. Next to relative inflows and outflows, as well as the past turnout rate, $T_{j,2019}$, still included in the vector $\mathbf{X_j}$, now the lagged turnout change ($\Delta T_{j,2019} \equiv T_{j,2019} - T_{j,2015}$) is also used to explain the dependent variable. As Figure 6 and Table B.2 in the Appendix document, distribution of flows is not orthogonal to past turnout trends. Strategic outflows are predominantly generated in big, high-turnout cities, with inflows concentrating in their metropolitan suburbs, structurally differing from more far-away locations. Specification (8) therefore asks whether the strategic inflows or outflows, given past turnout patterns, have yielded higher turnout growth than in case of other, unexposed units. In this way, the results can provide insight into the *reaction* of voters to flows between municipalities. Finally, the specification explicitly includes district-level fixed effects.

Table 3 reports the estimated model (8) in the second column. The first column shows coefficients of the model with the dependent variable replaced by an unadjusted turnout change ($\Delta T_j \equiv T_j - T_{j,2019}$) which includes both direct and behavioral effects. In this way, the two columns compare strength of effects with and without

the direct component. The last, third column presents turnout-growth acceleration relative to the previous electoral cycle, with $(\Delta T_j^{adj} - \Delta T_{j,2019})$ regressed on flows.

Table 3: Stage 2 dynamic results

	Dependent variable:		
	ΔT_j	ΔT_j^{adj}	$\Delta T_j^{adj} - \Delta T_{j,2019}$
	(1)	(2)	(3)
Inflow	0.2766***	0.1574**	0.1474***
	(0.0817)	(0.0773)	(0.0533)
Outflow	0.1059	0.2808	0.2335
	(0.2857)	(0.2740)	(0.1833)
Turnout19	−0.1903***	−0.1900***	
	(0.0176)	(0.0176)	
$\Delta T_{j,2019}$	−0.0059	−0.0064	
	(0.0349)	(0.0349)	
Control variables	Yes	Yes	Yes
District FE	Yes	Yes	Yes
Clustered SE (district)	Yes	Yes	Yes
Observations	2,429	2,429	2,429
Note:			*p<0.1; **p<0.05; ***p<0.01

As column (2) shows, only the inflows pattern survives when the pre-treatment turnout trends are taken into consideration. Relative to the past turnout rate and the past turnout change, municipalities exposed to inflows are expected to accelerate their turnout growth. Consistently with this result, high-inflow municipalities have raised their turnout between the elections of 2019 and 2023 more than between 2015 and 2019, as reported in column (3). At the same time, the inflow coefficient in column (2) is weaker than in column (1) but still statistically significant, suggesting relevance of both the mechanical and the behavioral effect for explaining voters' participation rates. However, no statistical significance at outflows coefficients suggests a correlational character of the respective results from Table 2. Persistently high-turnout areas are thus more likely to generate a surplus of voters eager to change their location but we no longer find evidence of local voters' response to the outflows. In other words, origins are endogenously selected among the municipalities with persistently high turnout and therefore higher potential for strategic outflows. Even if the pattern does not provide a robust empirical support for the theoretical Prediction

2 (mobilization in origin municipalities), it remains consistent with our theoretical framework. With higher turnout rates, effective vote power shrinks, increasing the relative profitability of moving to a different constituency, which leads to aggregate outflows. The results also gain additional credence with the wild bootstrap testing, with t-statistic 2.037 and p-value 0.028 at the inflow share coefficient from column (2) (2.764 and 0.047 respectively in column (3)). Additionally, Table B.3 in the Appendix documents the results' robustness to the alternative definition of treatment, as log of total flows.

Strategic inflows are naturally correlated with some characteristics of the municipalities, as voters flow mostly from urban centers to nearby suburban areas, whose turnout patterns may differ from municipalities located further away. However, given these turnout trends (past rate and its change), voters in municipalities exposed to inflows increase their turnout rate in a faster pace than elsewhere. Controlling for these trajectories helps account for pre-existing turnout dynamics differences across municipality types. At the same time, using turnout rates adjusted by the size of inflows disentangles the behavioral response from purely mechanical effects. The positive coefficient is consistent with local participation growing in municipalities exposed to higher inflows, after a correction by the direct contribution to turnout of incoming voters. Consistently with the destination mechanism in the model, the total inflows-turnout effect is determined both by the vote power channel and by the composition of incoming voters. The first of these channels is negative as it demobilizes local voters, with the latter being ambiguous and potentially mobilizing if it increases political competition. Positive net value of the coefficient can be then associated with mobilizing composition dominating the demobilizing vote power channel.

Overall, our results are thus consistent with the mild mobilization patterns in equilibrium. For origins, the pattern is purely descriptive. Origin municipalities tend to differ structurally from other locations, and outflows should therefore be treated as endogenous. Nevertheless, the finding aligns with our theoretical framework, in which high-turnout units are more likely to observe outflows due to decreasing vote power. In the dynamic setup, outflows no longer explain the turnout growth. For destinations, by contrast, the results provide stronger evidence of a local equilibrium response. Locations exposed to higher relative inflows exhibit faster growth of the adjusted turnout, consistently with local mobilization after the direct contribution of incoming voters is removed.

5 Conclusions

The paper studied presence and efficacy of voters' grassroots coordination by changing ballot location. Given frictions and obstacles to vote equality in many democratic systems, the question whether voters can counteract unfavorable institutional settings without formally changing them is highly relevant. A handy case of malapportionment resulting with strategic reallocation of voters across constituencies was provided by the 2023 Polish parliamentary elections. Given the limited administrative barriers to change the location of a ballot, evidence suggest that a substantial number of voters decided to cast their vote in a different location to increase their marginal impact on the outcome.

The theoretical framework presented in this paper explained a voter's decision to switch their voting location to maximize marginal impact on the electoral result. Despite spatial frictions that make voting outside the default constituency more costly, voters decided to re-register if the difference in their perceived margianl pivotality was sufficiently high. In response to the low-cost voters' re-registrations, also higher-cost voters, who otherwise abstain, may mobilize and turn out at the polls.

The empirical results presented in the paper are consistent with these predictions. The analysis indicated that a significant portion of voters who changed the location of their ballot ahead of the parliamentary elections did it for strategic reasons. Voters' inflows are strongly related to the vote power and race closeness in their neighboring constituencies. Using 2023 Polish parliamentary election data, we estimated the size of the strategic mobility among voters and evaluated its relationship to the turnout equilibrium response conditioning on the past trends of turnout changes. The results are consistent with mild local mobilization among local voters in targeted municipalities, as the adjusted turnout in municipalities with higher exposure to strategic inflows grew faster given past turnout levels and dynamics. While the relationship between the inflows and local adjusted turnout in destinations, conditional on past trends, provides evidence consistent with the theoretical predictions, an analogous relationship for origin municipalities was not robust to pre-existing turnout patterns.

In terms of the theoretical framework, future work could incorporate more than two locations. Although the predictions already inform the empirical setup, considering multiple locations may provide additional insights into the trade-off between marginal pivotality gain and the cost of travel. The identification of mobilization patterns could benefit from more granular, individual-level data on origin-destination

voters' preferences, location choices and willingness to vote. Such data, if available, could also help disentangle the vote-power and race-closeness channels underlying the turnout response. Finally, a natural extension of the empirical framework would be to structurally estimate the parameters governing the costs of participation and preferences for re-registration, in the spirit of (Kawai, Toyama, and Watanabe, 2021) and (Berry, Cox, and Haile, 2025). Such an exercise could recover the relative importance of electoral incentives and mobility costs in voters' location choices. This would allow for a better understanding of the underlying preferences and frictions driving the electoral mobility across different constituencies.

The paper provides insights into decisions of voters in a rare strategic setting. The combination of malapportionment and the ease of changing one's ballot location allowed voters to flow between constituencies to increase their marginal perceived pivotality. It resulted in a significant number of voters choosing a different voting location. The institutional setting of Polish parliamentary elections provided a novel context to study voters' choices and expanded the usual question, whether to vote or abstain, to a third option of voting elsewhere. If appropriately coordinated, the estimated number of strategic voters was large enough to effectively alter the allocation of several marginal parliamentary seats, highlighting the relevance of a substantial grassroots response to the malapportionment.

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A Appendix

A Proofs for the theoretical model

Proof of Lemma 1. Consider first a voter assigned to constituency B. Voting at home gives

$$U_i(H) = q_B - c_i,$$

while re-registering and voting in A gives

$$U_i(T) = q_A - \alpha c_i.$$

The utility gain from voting at home rather than traveling is

$$U_i(H) - U_i(T) = q_B - q_A + (\alpha - 1)c_i.$$

Suppose that $q_B \geq q_A$, then this expression is strictly positive for every $c_i > 0$, since $\alpha > 1$. Hence traveling from B to A is strictly dominated by voting at home. Under the tie-breaking rule, a voter also does not travel when indifferent. No voter from B chooses to travel to constituency A.

Now consider a voter assigned to constituency A. The utility gain from traveling to B rather than voting at home is

$$U_i(T) - U_i(H) = q_B - q_A - (\alpha - 1)c_i. \tag{9}$$

For any q_B such that $q_B \leq q_A$, this expression is weakly negative for $c_i = 0$ and strictly negative for every $c_i > 0$. Under the tie-breaking rule, no voter travels when indifferent. This implies the following:

$$q_B \leq q_A \implies M_{(A,B)}(s, \omega) = 0.$$

If $q_B > q_A$, the gain in (9) is strictly positive for all

$$c_i < \frac{q_B - q_A}{\alpha - 1},$$

and traveling also strictly dominates abstention for all $c_i < q_B/\alpha$. The set of costs for which T is strictly optimal is therefore the interval $\big[0, \min\{(q_B - q_A)/(\alpha - 1),\, q_B/\alpha\}\big)$, which has positive length. Since F admits a strictly positive density on its support, and $F(0) = 0$, the interval has a strictly positive length, so a strictly positive flow of voters from A to B satisfies individual optimality whenever $q_B > q_A$.

□

Proof of Proposition 1. Fix a flow M such that

$$\Delta q(M) = q_B(M) - q_A(M) > 0.$$

To simplify notation, write

$$q_A = q_A(M), \qquad q_B = q_B(M).$$

For a voter assigned to constituency A, the utilities are

$$U_i(0) = 0, \qquad U_i(H) = q_A - c_i, \qquad U_i(T) = q_B - \alpha c_i.$$

First, compare traveling with voting at home:

$$U_i(T) > U_i(H) \iff q_B - \alpha c_i > q_A - c_i \iff c_i < \frac{q_B - q_A}{\alpha - 1} = c_{TH}(M).$$

Second, compare traveling with abstention:

$$U_i(T) > U_i(0) \iff q_B - \alpha c_i > 0 \iff c_i < \frac{q_B}{\alpha} = c_{T0}(M).$$

Under the tie-breaking rule, a voter travels only when traveling is strictly preferred to both alternatives. Therefore, voters assigned to A travel to B if

$$c_i < \bar{c}_T(M),$$

where

$$\bar{c}_T(M) = \min\{c_{TH}(M), c_{T0}(M)\}.$$

Now consider voting at home. Voting at home weakly dominates abstention whenever

$$U_i(H) \geq U_i(0) \iff q_A - c_i \geq 0 \iff c_i \leq q_A = c_{H0}(M).$$

Voting at home weakly dominates traveling whenever

$$U_i(H) \geq U_i(T) \iff q_A - c_i \geq q_B - \alpha c_i \iff c_i \geq \frac{q_B - q_A}{\alpha - 1} = c_{TH}(M).$$

Hence voters assigned to A vote at home whenever

$$c_{TH}(M) \leq c_i \leq c_{H0}(M).$$

It remains to verify that all remaining voters assigned to A abstain. If

$$c_i < c_{TH}(M)$$

but the voter does not travel, then it must be that

$$c_i \geq c_{T0}(M),$$

so traveling does not dominate abstention, while traveling dominates home. Hence abstention is optimal. If instead

$$c_i > c_{H0}(M),$$

then home voting is worse than abstention. If $c_i \geq c_{TH}(M)$, home weakly dominates traveling, so abstention also dominates traveling. If, as shown, above, $c_i < c_{TH}(M)$, then traveling dominates home, but since the voter does not travel it must hold that $c_i \geq c_{T0}(M)$, so abstention weakly dominates traveling. Thus for all the remaining cost intervals, voters do not vote home nor travel, but abstain.

Finally, consider voters assigned to constituency B. Since $\Delta q(M) > 0$, Lemma 1 implies that they do not travel to A. They therefore choose between voting at home and abstaining. Voting at home is weakly optimal whenever

$$q_B - c_i \geq 0,$$

or equivalently

$$c_i \leq q_B(M).$$

Thus,

$$a_i^{BR}(B, c_i; M) = \begin{cases} H, & \text{if } c_i \leq q_B(M), \\ 0, & \text{otherwise.} \end{cases}$$

This proves the cutoff rule. □

Proof of Proposition 2. Throughout, maintain the assumptions that F is continuous with $F(0) = 0$, and that $M \mapsto v_j(M)$ and $M \mapsto d_j(M)$ are continuous for $j \in \{A, B\}$. Since q is continuous, $q_A(M)$, $q_B(M)$, and

$$\Delta q(M) = q_B(M) - q_A(M)$$

are continuous on $[0, N_A]$.

First, let us start by proving that Φ is well-defined and maps $[0, N_A]$ into itself. By Proposition 1, given a candidate flow M with $\Delta q(M) > 0$, the mass of voters assigned to A whose best response is T equals

$$N_A F(\bar{c}_T(M)).$$

If $\Delta q(M) \leq 0$, Lemma 1 implies that this mass is zero. Hence

$$\Phi(M) = \begin{cases} N_A F(\bar{c}_T(M)), & \text{if } \Delta q(M) > 0, \\ 0, & \text{if } \Delta q(M) \leq 0. \end{cases}$$

Since $0 \leq F(\cdot) \leq 1$, we have $\Phi(M) \in [0, N_A]$ for every $M \in [0, N_A]$.

Second, note that Φ is continuous on $[0, N_A]$. On the set where $\Delta q(M) > 0$, the cutoffs

$$c_{TH}(M) = \frac{\Delta q(M)}{\alpha - 1}, \qquad c_{T0}(M) = \frac{q_B(M)}{\alpha}$$

are continuous, and therefore so are

$$\bar{c}_T(M) = \min\{c_{TH}(M), c_{T0}(M)\}$$

and

$$\Phi(M) = N_A F(\bar{c}_T(M)).$$

On the set where $\Delta q(M) < 0$, $\Phi(M) = 0$, so continuity is immediate. It remains to check continuity at any point M_0 such that $\Delta q(M_0) = 0$. Along any sequence $M_n \to M_0$ with $\Delta q(M_n) > 0$,

$$0 \leq \bar{c}_T(M_n) \leq c_{TH}(M_n) = \frac{\Delta q(M_n)}{\alpha - 1} \to 0.$$

Thus $\bar{c}_T(M_n) \to 0$. Since F is continuous and $F(0) = 0$,

$$\Phi(M_n) = N_A F(\bar{c}_T(M_n)) \to 0 = \Phi(M_0).$$

Along sequences with $\Delta q(M_n) \leq 0$, $\Phi(M_n) = 0 = \Phi(M_0)$. Hence Φ is continuous at M_0, and therefore on all of $[0, N_A]$.

Third, let us see that an equilibrium flow exists. Define

$$g(M) = M - \Phi(M).$$

The function g is continuous on $[0, N_A]$. Moreover,

$$g(0) = -\Phi(0) \leq 0,$$

and

$$g(N_A) = N_A - \Phi(N_A) \geq 0,$$

because $\Phi(N_A) \leq N_A$. By the intermediate value theorem, there exists $M^* \in [0, N_A]$ such that

$$g(M^*) = 0.$$

Equivalently,

$$M^* = \Phi(M^*).$$

Fourth, let us show that party-specific flows preserve the initial party composition in the origin. If $M^* = 0$, party-specific flows are zero and the claim is immediate. If $M^* > 0$, then $\Delta q(M^*) > 0$ and

$$M^* = N_A F(\bar{c}_T(M^*)).$$

Hence

$$F(\bar{c}_T(M^*)) = \frac{M^*}{N_A}.$$

Because the cutoff rule is common to supporters of both parties and the cost distribution does not vary with party type,

$$M^*_{(A,B),L} = \pi_{AL} F(\bar{c}_T(M^*)) = N_A \gamma_A \frac{M^*}{N_A} = \gamma_A M^*,$$

and

$$M^*_{(A,B),R} = \pi_{AR} F(\bar{c}_T(M^*)) = N_A (1 - \gamma_A) \frac{M^*}{N_A} = (1 - \gamma_A) M^*.$$

Finally, it remains to prove the minimal equilibrium flow exists. The set of equilibrium flows is

$$\mathcal{E} = \{M \in [0, N_A] : M = \Phi(M)\}.$$

Equivalently,

$$\mathcal{E} = g^{-1}(\{0\}).$$

Since g is continuous, $\mathcal{E}$ is closed. By the existence result above, it is non-empty. Since $\mathcal{E} \subset [0, N_A]$, it is compact. Therefore,

$$M^{\min} = \min \mathcal{E}$$

exists. □

B Tables and figures for empirical analysis

Interest in changing voting location in time: As the figure below illustrates, the pre-election Google searches of the phrase "Change of voting location" peaked around ten times higher in 2023 than in 2019. Intriguingly, the 2019 interest in changing location seemed even lower than in the 2020 presidential election, where, by design, voting location plays no role for the results. However, the presidential election took place during the summer season (28.06.2020 and 12.07.2020).

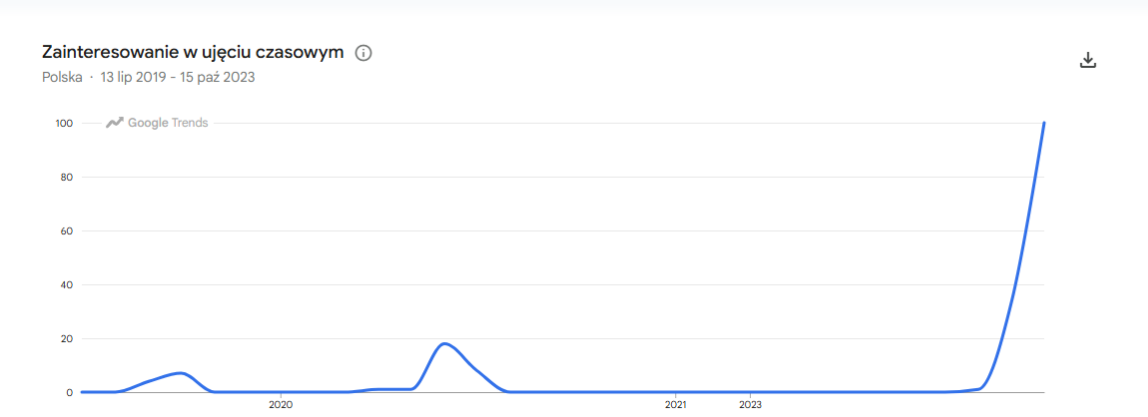


Figure 5: Google Trends searches for the phrase "Zmiana miejsca głosowania" ("Change of voting location") in 2019 (left) and 2023 (right).

Stage 1 additional distance thresholds: Table B.1 compares different distance thresholds for the neighbors' list creation in the interaction specification. Increasing the threshold from 30km to 50km has virtually no effect on the coefficient estimates, with (lagged) border and lagged vote power variables gradually gaining significance in higher thresholds.

Turnout trends and Stage 2 treatment: Figure 6 and Table B.2 motivate the transition from the early model (7) reported in Table 2 to the dynamic model (8) from Table 3. The figure illustrates the pre-treatment turnout patterns. High-turnout areas clearly cluster in big cities and their suburbs, possibly confounding a relationship between the flows and the turnout response. Voters mobility thus should not be expected to be orthogonal to past turnout trends, justifying the shift to a dynamic (turnout change) setting, with flows' size conditional on these very trends.

Table B.1: Stage 1 alternative thresholds

	Dependent variable: Add23			
	30km	50km	70km	100km
	(1)	(2)	(3)	(4)
Add19	0.879***	0.892***	0.898***	0.905***
	(0.017)	(0.016)	(0.016)	(0.016)
VP	0.015	0.008	−0.001	0.001
	(0.012)	(0.011)	(0.009)	(0.009)
Border	0.001	0.001	0.002	0.002*
	(0.001)	(0.001)	(0.001)	(0.001)
Close	−0.241***	−0.196***	−0.157***	−0.120***
	(0.024)	(0.021)	(0.018)	(0.017)
Close × VP	0.231***	0.188***	0.150***	0.115***
	(0.023)	(0.020)	(0.018)	(0.016)
lag_Add19	0.037	0.037	0.075	0.029
	(0.040)	(0.061)	(0.088)	(0.132)
lag_VP	0.003	0.023	0.058**	0.078**
	(0.016)	(0.017)	(0.020)	(0.027)
lag_Border	−0.004	−0.005	−0.011*	−0.017**
	(0.002)	(0.003)	(0.005)	(0.007)
lag_Close	0.357***	0.363***	0.376***	0.464***
	(0.030)	(0.032)	(0.036)	(0.050)
lag(Close × VP)	−0.341***	−0.347***	−0.359***	−0.445***
	(0.029)	(0.031)	(0.035)	(0.048)
Controls	yes	yes	yes	yes
Observations	2429	2429	2429	2429
Note:				*p<0.05; **p<0.01; ***p<0.001

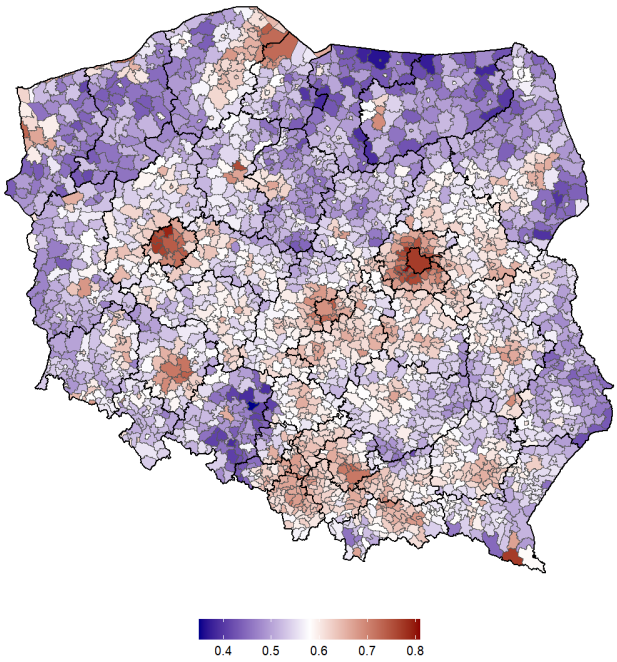


Figure 6: Turnout in 2019 (share of eligible voters)

Figure 6 strengthens this choice, by showing that the 2023 inflows are good predictors of the turnout rate already in 2019, even for highest-turnout municipalities. Inflows concentrate around big cities, in municipalities structurally different from less suburban locations.

The applied Propensity Score Matching exercise facilitates a close comparison of units across the sample by matching high-inflow destinations with observationally similar municipalities. We define a binary treatment *Destination* taking value 1 for destinations in the k-th percentile of relative inflows (and 0 otherwise) ($k \in \{95, 97, 98, 99\}$), and test whether the treatment group obtains higher turnout rates already in 2019. We use a nearest neighbor 1-1 matching (without replacement) based on a logistic propensity score. Covariates used in the propensity score computation include original control variables lagged in values, with standard errors clustered at the district level. For all percentiles except the top one (where the sample becomes tiny), future inflows predict current turnout well.

Table B.2: PSM Destination – full sample, lagged

	Dependent variable: T19			
	Top 1%	Top 2%	Top 3%	Top 5%
	(1)	(2)	(3)	(4)
Destination	0.0022	0.0125**	0.0080**	0.0073***
	(0.0046)	(0.0051)	(0.0036)	(0.0028)
Turnout15	0.9246***	0.7955***	0.7859***	0.7997***
	(0.0641)	(0.0553)	(0.0503)	(0.0233)
Threshold	0.0184	0.0116	0.0092	0.0058
Observations	50	98	146	244
Note:				*p<0.1; **p<0.05; ***p<0.01

Alternative treatment definition in Stage 2 dynamic model: Table B.3 illustrates the results of the dynamic model (8) (columns (2) in Table 3) with an alternative treatment definition. Instead of flow shares, it takes flow logs:

$$Log(inflow) = \begin{cases} \log(Add_diff) & \text{if } Add_diff > 0 \\ 0 & \text{otherwise} \end{cases} \tag{10}$$

$$Log(outflow) = \begin{cases} \log(-Add_diff) & \text{if } Add_diff < 0 \\ 0 & \text{otherwise.} \end{cases} \tag{11}$$

As depicted in the table, the results are robust to this change.

Table B.3: Stage 2 results

	Dependent variable:
	Change in turnout Dynamic
Log(inflow)	0.0014**
	(0.0007)
Log(outflow)	0.0008
	(0.0005)
Turnout19	−0.1904***
	(0.0176)
$\Delta T_{j,2019}$	−0.0056
	(0.0357)
Control variables	Yes
District FE	Yes
Clustered SE (district)	Yes
Observations	2,429
R^2	0.5550
Adjusted R^2	0.5441
Note:	*p<0.1; **p<0.05; ***p<0.01